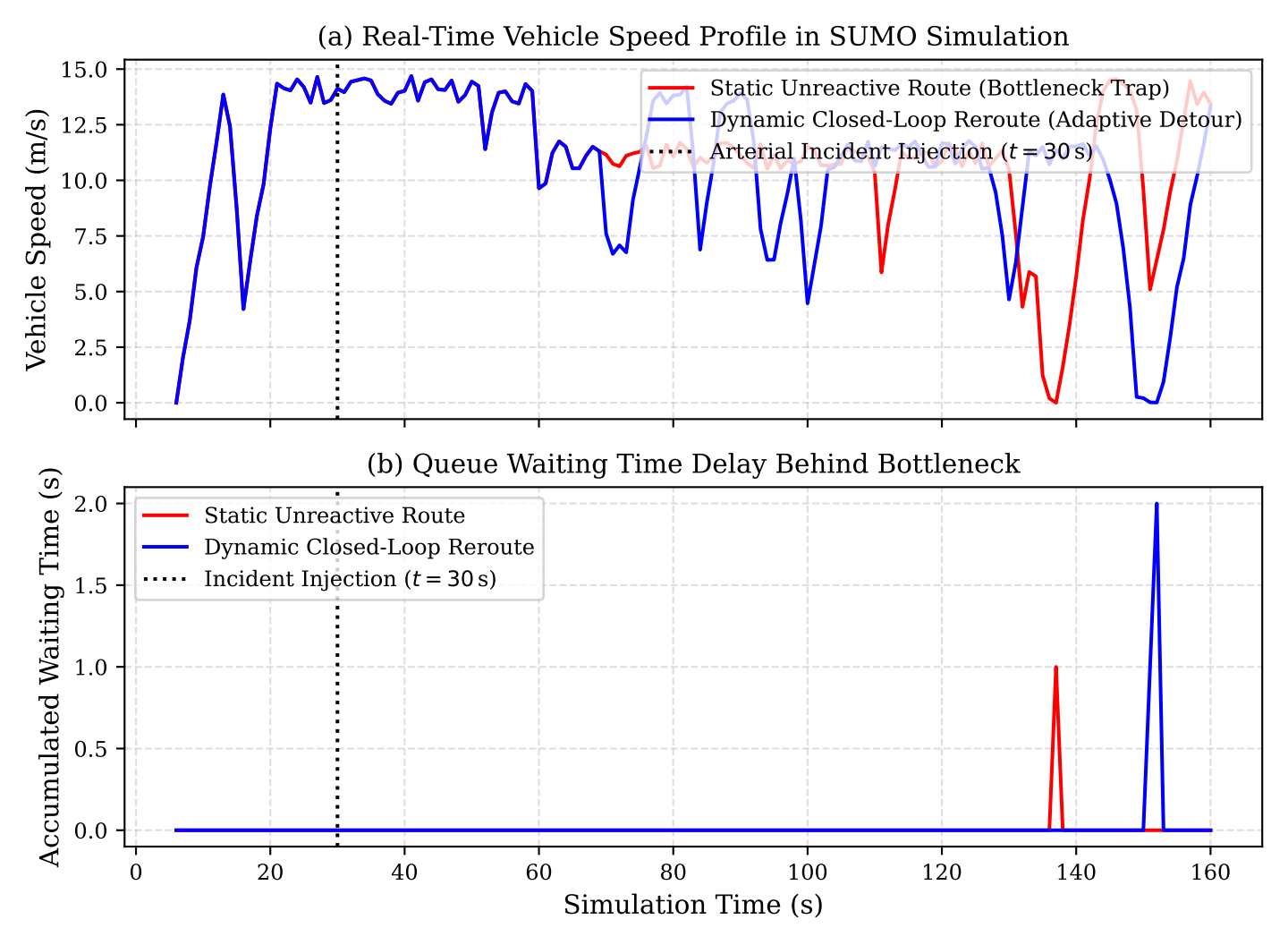
(a) Real-Time Vehicle Speed Profile in SUMO Simulation
Vehicle Speed (m/s)
15.0
12.5
10.0
7.5
5.0
2.5
0.0
Static Unreactive Route (Bottleneck Trap)
Dynamic Closed-Loop Reroute (Adaptive Detour)
Arterial Incident Injection ($t = 30$ s)
(b) Queue Waiting Time Delay Behind Bottleneck
Accumulated Waiting Time (s)
2.0
1.5
1.0
0.5
0.0
Static Unreactive Route
Dynamic Closed-Loop Reroute
Incident Injection ($t = 30$ s)
0
20
40
60
80
100
120
140
160
Simulation Time (s)